

A First Look at Defense

Packages, the box, man and zone, one-high and two-high, and the blitz: the five things to read about a defense before the snap, and why each one exists.

Second-and-7, early in the second quarter. The offense jogs out of the huddle with three wide receivers, and before the quarterback has his hands under the center the analyst has already said three things: “They stay in their nickel. Six in the box. Two high.” Then the ball is snapped, a linebacker who looked like he was going to rush the passer turns and runs backward, and the quarterback throws the ball away. “They showed blitz and dropped out of it.” To a casual fan that is a stream of jargon. To the analyst it is a quick inventory: *who* the defense sent onto the field, *how many* of them are waiting for a run, *how many* are guarding the deep part of the field, and *how many* came after the quarterback. That inventory is what this chapter teaches. It is a first look, deliberately shallow, so the chapters on the run and pass games can talk about defenders by name. The full treatment of each idea comes later: fronts in Part 5, coverages in Part 6, pressure in Part 7.

You’ll be able to...

- Name the common defensive packages (base 4-3 and 3-4, nickel, dime, quarter, big nickel, goal line), say which offensive personnel each one is meant to meet, and explain where “nickel” and “dime” got their names.
- Count the box on a broadcast, compare it with the offense’s blockers, and say whether the numbers favor a run; point out the apex or overhang defender who stands between the box and a slot receiver.
- Say what man and zone coverage are, and what each is good at and bad at.
- Tell a one-high shell from a two-high shell before the snap, and say what each protects and what it gives away.
- Count rushers after the snap and call a four-man rush or a blitz.
- Explain why nickel, not the 4-3 or the 3-4, is the modern NFL’s real base defense, and why that changed a little in 2025.

The defense’s problem: eleven players against a play it hasn’t seen

Start with the asymmetry. The offense knows what it is going to do. Its play-caller chose a play, the quarterback told ten teammates, and every one of them knows where the ball is supposed to go (What a Play Really Is). The defense does not. Its coordinator called his own play at the same moment, blind, and everything after that is reaction. A defense is eleven people trying to be ready for a run in either direction, a short pass, a long pass and a quarterback scramble, all at once, without knowing which.

They cannot be ready for everything equally, because there are only eleven of them. Every defender placed near the ball to stop a run is a defender not standing 15 yards deep to stop a long pass. Every defender sent after the quarterback is one fewer covering a receiver. So a defensive call is a set of bets about where to spend eleven players, and almost everything you can read about a defense before the snap is a clue to how those players were spent. This chapter breaks the spending into five questions, in the order you can answer them on a broadcast:

1. **Who is on the field?** The package: how many big run-stoppers, how many fast pass defenders.
2. **How many are near the ball?** The box count, compared with the offense’s blockers.
3. **Who is stuck in between?** The apex or overhang defender, half run-stopper, half pass defender.
4. **How will they guard receivers?** Man or zone, and one deep safety or two (the shell).
5. **How many will rush the passer?** Four, or five and more (a blitz).

A real defensive call bundles the answers into a few words: a word for the **front** (where the linemen and linebackers line up; Part 5), a word or number for the **coverage** (how the players behind them split up the receivers; Part 6), and sometimes a **pressure** (who rushes; Part 7). The package is not even in the call. It is decided a few seconds earlier, when players run on and off the field.

Packages: who comes onto the field

A **defensive package** is the group of eleven defenders on the field for a given snap. Football players are specialists: a 320-pound defensive tackle and a 190-pound cornerback are both “defense,” but they are built for opposite jobs (The Twenty-Two walks through every position). A package is the defense’s choice of how many of each body type to use, and teams name packages mostly by counting one group: the **defensive backs** (DBs), the cornerbacks and safeties of the secondary. Linemen and linebackers make up the rest, and since there are always eleven, more DBs means fewer big players.

Base: the 4-3 and the 3-4

A **base defense** is the traditional starting package: four defensive backs and seven bigger players in the front seven. There are two classic ways to split those seven, and their names are simply the counts:

- The **4-3**: four defensive linemen and three linebackers (the Sam, Mike and Will of The Twenty-Two).
- The **3-4**: three defensive linemen and four linebackers. The two outside linebackers often stand on the line of scrimmage, at the ends of the line, and on most pass plays they rush the quarterback like defensive ends.

Both were built for an older offense, one with two running backs or two tight ends, where the main threat was a run and the third wide receiver did not exist. The 4-3 became the NFL's standard in the late 1950s; it is usually credited to Tom Landry, the New York Giants' defensive coordinator, who made it the Giants' base defense in 1956, and after the Giants won the championship that year much of the league copied it.¹ The 3-4 grew out of the college "Oklahoma 5-2" of Bud Wilkinson; in the NFL, Chuck Fairbanks's 1974 New England Patriots are usually named as the first team to play it regularly, and it was the league's dominant base defense around 1980.²

For a first look, the difference that matters is smaller than the names suggest. In both, the defense usually rushes **four** players on a pass (the 3-4's fourth rusher is an outside linebacker standing up instead of a lineman with his hand on the ground), and both put seven players near the ball against the run. What really separates them, which linemen take on which blockers and who is responsible for which gap, is the subject of Even Fronts and Odd and Hybrid Fronts, and Appendix C has a one-page comparison card. For now: **the digits count linemen and linebackers, and the base package always has four defensive backs.**

Common misconception

"A 3-4 defense rushes three." Almost never on a passing down. The outside linebackers in a 3-4 are usually the team's best pass rushers (they are edge rushers who happen to be listed as linebackers), so a 3-4 team rushing four, with one of the outside linebackers dropping into coverage or both rushing and one lineman dropping, is the normal case. The labels describe how the defense lines up, not how many come after the quarterback. Count rushers after the snap instead (the last section of this chapter).

Nickel and dime: taking linebackers off the field

The trouble with base defense starts with the third wide receiver. Against an offense with three receivers, a base defense has four DBs for three receivers plus a tight end and a back who can also run routes. Someone from the front seven, usually a linebacker, has to line up across from the third receiver, the one in the **slot** (the gap between the offensive line and a wide receiver). That linebacker might weigh 240 pounds; the slot receiver might run the 40-yard dash a few tenths of a second faster. That is a mismatch the offense will find on every snap.

The answer is a substitution. Take a linebacker off the field and send in a fifth defensive back. That package is the **nickel**, and the fifth DB, usually a cornerback who specializes in covering slot receivers, is the nickel back or slot cornerback. Take out a second linebacker for a sixth DB and you have the **dime**. The names are coins: a nickel is five cents, so the fifth DB is "the nickel," and a dime is two nickels, so the sixth is "the dime."

¹Landry's role: Pro Football Hall of Fame, Tom Landry ("a player-coach in 1954 and 1955 before becoming a full-time defensive coach from 1956 to 1959"): <https://www.profootballhof.com/players/tom-landry/>. The 4-3 as the Giants' base in 1956 and its spread after the 1956 title: Wikipedia, "4-3 defense" ("In 1956, Landry became the first defensive coordinator to switch to the 4-3 as a base defense"; the article also traces it to Steve Owen's umbrella defense of 1950): https://en.wikipedia.org/wiki/4%E2%80%933_defense; Wikipedia, "Sam Huff" (Landry built the new 4-3 around the rookie middle linebacker in 1956): https://en.wikipedia.org/wiki/Sam_Huff. Credit for the 4-3 is, like most scheme origins, shared and debated, hence "usually credited."

²Wikipedia, "3-4 defense" (Wilkinson's Oklahoma 5-2; the 1974 Patriots under Chuck Fairbanks as the first NFL team to use it regularly; its peak in the late 1970s and early 1980s): https://en.wikipedia.org/wiki/3%E2%80%934_defense.

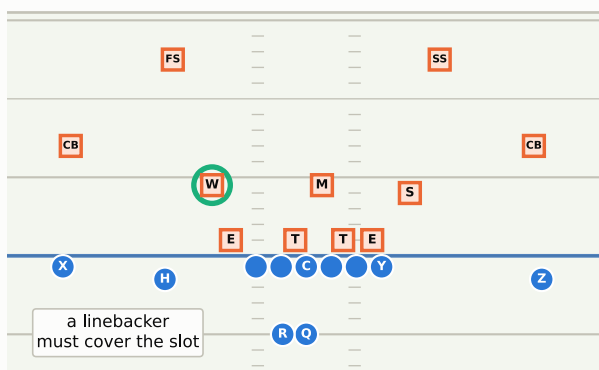
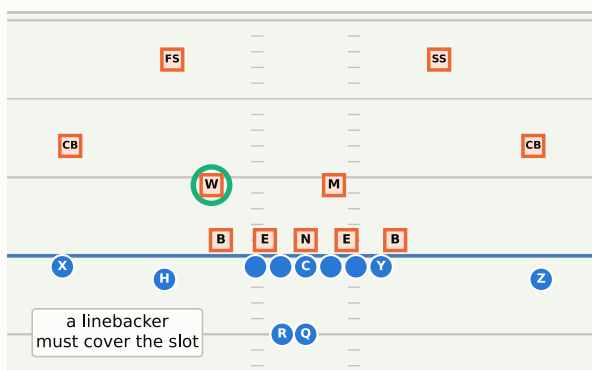
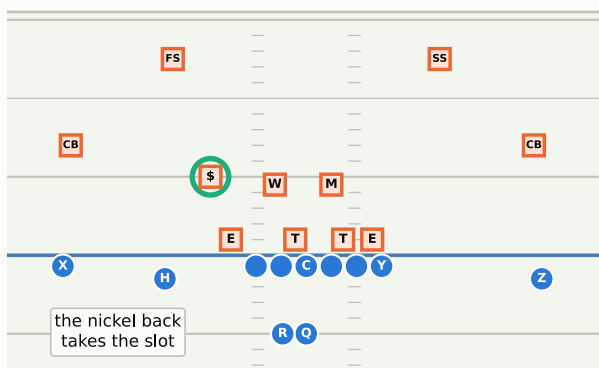
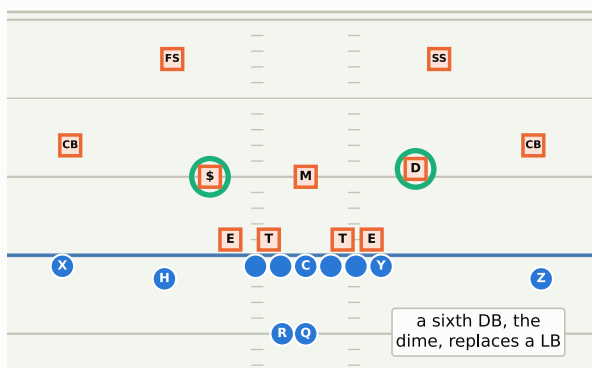
Base 4-3: 4 DL, 3 LB, 4 DB**Base 3-4: 3 DL, 4 LB, 4 DB****Nickel: 4 DL, 2 LB, 5 DB****Dime: 4 DL, 1 LB, 6 DB**

Figure 1: The same offense (11 personnel: one back, one tight end, three wide receivers) against four packages. In both base defenses (top) the four defensive backs are already spent on the two outside receivers and the two deep safeties, so a linebacker (ringed) has to walk out over the slot receiver, H. Nickel (bottom left) replaces that linebacker with a fifth defensive back (\$, the nickel back). Dime (bottom right) replaces a second linebacker with a sixth (D, the dime back), leaving one linebacker (M). Labels: E = defensive end, T = defensive tackle, N = nose tackle, B = outside linebacker standing on the line, S/M/W = Sam, Mike and Will linebackers, CB = cornerback, FS/SS = safeties.

The idea is older than most fans assume. The usual account credits Jerry Williams, a Philadelphia Eagles assistant, with putting a fifth defensive back on the field around 1960, and George Allen, then a Chicago Bears assistant, with giving it the name “nickel” (and, by some accounts, with taking the credit). Don Shula and his defensive coordinator Bill Arnsparger made nickel and dime packages a routine part of the Miami Dolphins’ defense in the 1970s.³ As with most “who invented it” stories in football, treat the details as contested; the coin joke has outlived everyone’s claim to it.

The packages keep going in both directions. A table is the quickest way to hold them:

³Wikipedia, “Nickel defense” (Jerry Williams, Philadelphia, 1960; George Allen and the name; Shula and Arnsparger’s Dolphins in the 1970s; Arnsparger’s dime sets): https://en.wikipedia.org/wiki/Nickel_defense. On the coin names (a dime is “the next step up” from a nickel; “quarter” packages beyond it): Wikipedia, “Dime defense”: https://en.wikipedia.org/wiki/Dime_defense.

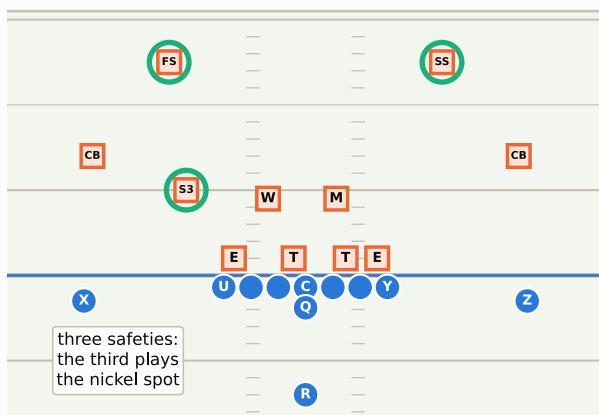
Table 1: Defensive packages from heaviest to lightest. Personnel numbers (21, 12, 11) are the offense's backs-then-tight-ends code from Personnel; the DL-LB splits vary by team.

Package	Typical make-up (DL-LB-DB)	Usually meets	What it is for
Goal line	6-3-2 or 5-3-3	23 or jumbo, inside the 3	Fill every gap
Base 4-3 / 3-4	4-3-4 / 3-4-4	21, 12, 22 personnel	Stop the run, cover a tight end
Big nickel	4-2-5, three safeties	12, or 11 with a big tight end	Nickel speed, base size
Nickel	4-2-5 (or 3-3-5, 2-4-5)	11 personnel	Three receivers: the default
Dime	4-1-6 (or 3-2-6)	10, empty, third-and-long	Four or five receivers
Quarter (dollar)	3-1-7 or 4-0-7	Last-play heaves	Nothing gets behind anyone

The **quarter** package adds a seventh DB: four quarters in a dollar, and some staffs do call it the **dollar**, though the coin names get loose at this end of the scale and teams use them differently. You will see seven DBs mostly on the last play of a half, when the offense is about to throw the ball as high and far as it can.

At the other end, the **goal-line** package is all size: six or more linemen, three linebackers and as few as two DBs, for the snap when the offense needs one or two yards and everybody in the stadium knows it may run (how those snaps are defended is the goal-line defense of Part 10). And in between sits a modern hybrid, the **big nickel**: a nickel whose fifth defensive back is a third safety rather than a cornerback. Safeties are usually bigger and better tacklers than corners, so a big nickel can stay on the field against a two-tight-end offense without getting pushed around, and still cover like a nickel if the offense throws.

Big nickel vs 12 personnel



Goal line (6-3-2) vs 23 personnel, at the 2

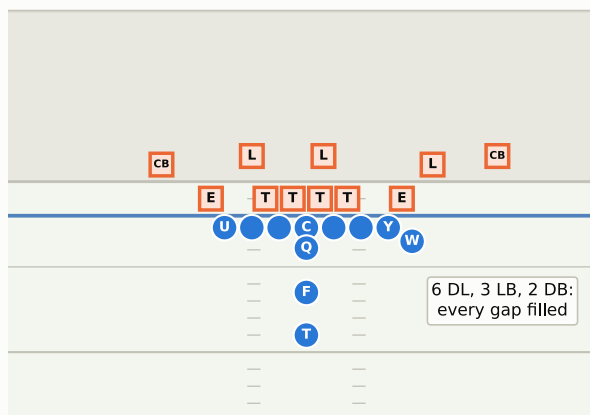


Figure 2: Two packages for bigger offenses. Left: big nickel against 12 personnel (two tight ends, U and Y). It is still a five-defensive-back package, but the fifth is a third safety (S3, ringed with the two deep safeties), here walked down outside the left tight end where a linebacker would stand in base. Right: a 6-3-2 goal-line package against 23 personnel (two backs and three tight ends, one of them, W, off the line as a wing) at the 2-yard line: six linemen covering every gap, three linebackers (L), two cornerbacks. The grey area is the end zone; the offense's fullback (F) and tailback (T) are stacked behind the quarterback.

Why packages exist: the defense gets to answer the substitution

Why not just pick the best eleven and leave them out there? Because the offense changes its players first, and the defense gets to answer. The offense decides its personnel before each snap; when it substitutes, the officials give the defense time to send in matching players (substitution matching, covered in Personnel). So the package is the defense's reply to the offense's personnel: three wide receivers come on, the nickel back runs on and a linebacker runs off. A defensive staff has a person on the sideline whose job during the game is mostly this: watch who the offense sends in and shout the matching package.

Every package is a trade of size for speed. Nickel covers three receivers well and is one linebacker lighter against a run. Dime covers the field and is easy to run on. Goal line stuffs the middle and has almost nobody to cover a pass. Offenses know it, which is why personnel is a weapon: line up with players who can both block and catch (a tight end who runs like a receiver, a fullback who catches passes), and the defense has to choose which weakness to accept. That tug of war is the thread of this whole course.

i Madden vs. real life

In Madden you pick “Nickel 2-4-5” from a menu, and the package, the alignment and the coverage arrive together. In a real game they are three separate decisions made at different moments. The **package** is a substitution, made in the few seconds after the offense’s players run on, usually by a coach shouting names on the sideline. The **front** and **coverage** come in the defensive play call, through the one defender with a helmet radio (the green dot). And the offense can still undo some of it: an offense that hurries to the line without substituting can freeze the defense in a package it would rather not be in. You will meet that tactic in In-Game Adjustments and Tempo.

Why nickel is the modern base defense

Here is the twist that makes “base defense” a slightly misleading phrase. In today’s NFL the base package is not the one teams use most. Offenses play 11 personnel (one back, one tight end, three receivers) on roughly 60 percent of their snaps,⁴ and three receivers call for nickel. So nickel has become what base used to be: the package a defense plays most, the one its best eleven players are built around. Many teams count their nickel back as a starter and pay him like one.

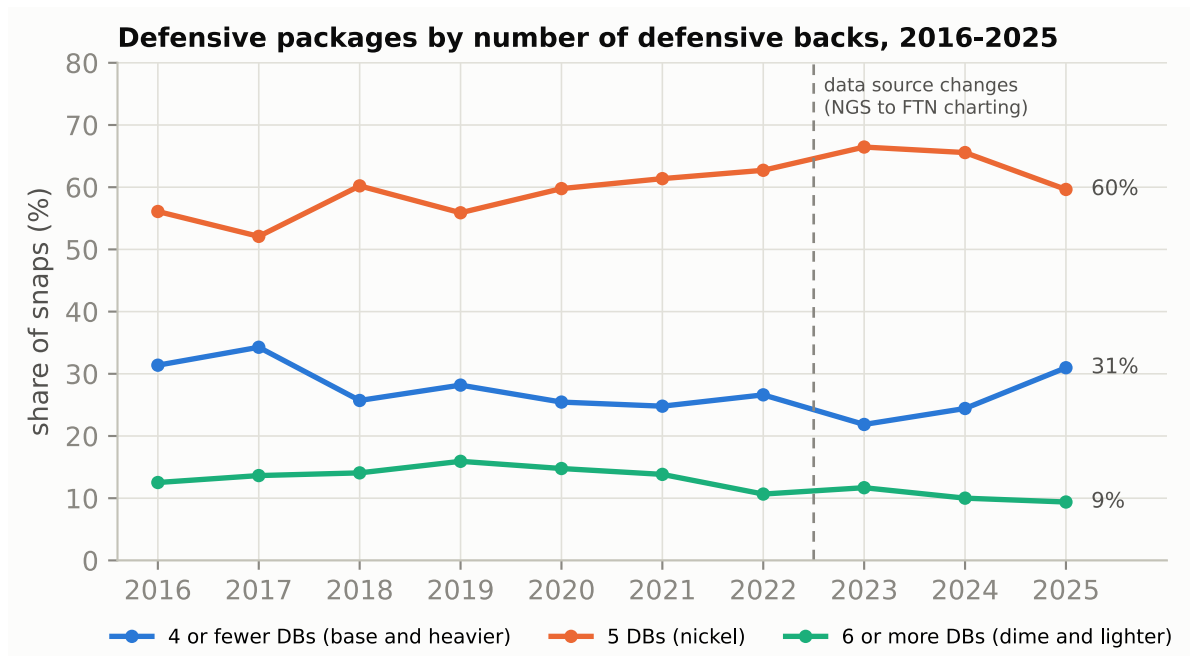


Figure 3: How often defenses use each package, regular seasons 2016-2025, grouped by the number of defensive backs on the field (all run and pass plays with personnel recorded). Five defensive backs (nickel) has been the most common package every season; four or fewer (base and heavier) bottomed out in 2023 and rebounded in 2025, when offenses used more two-tight-end sets. Data: nflverse participation data via nflreadpy; 2016-2022 from NGS, 2023-2025 from FTN charting, which lists positions differently (dashed line), so read across that line with care.

The chart tells two stories. The first is the steady one: in every season since 2016, five defensive backs was the most common look, and five or more was on the field for roughly two-thirds to three-quarters of all snaps.⁵ The second is the recent wobble. Base and heavier packages fell to about one snap in five in 2023 and then climbed back to about three in ten in 2025. NFL Next Gen Stats, measuring a little differently, has the same shape: defenses played base on 29.7 percent of plays in 2025, after a low of 20.5 percent in 2023 and no season above 25 percent in the five before.⁶

⁴This book’s calculation from nflverse participation data, regular-season run and pass plays, computed the same way as the package chart below: 11 personnel was between 58.8 and 64.6 percent of snaps in every season 2016-2025, with the decade low (58.8) in 2025. nflverse in Python shows how to reproduce it.

⁵This book’s calculation, the chart above: regular-season run and pass plays with defensive personnel recorded, about 32,000 to 34,000 a season. Defensive backs are counted from the DB field through 2022 and from the CB, FS and SS fields from 2023. Because FTN lists roster positions, a safety who plays linebacker still counts as a defensive back from 2023 on; goal-line packages with three or fewer DBs (one to two percent of snaps) are grouped with base.

⁶NFL Next Gen Stats Analytics Team, “NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks,” NFL.com (2026): “the prevalence of heavy personnel induced defenses to play base on 29.7% of plays in 2025 after they had not done so at a rate above

The reason is the offenses. In 2025, 11 personnel fell to its lowest share of the decade and 12 personnel (two tight ends) rose to its highest (Personnel has the numbers). Two tight ends is a direct attack on nickel: if the defense stays small, the offense runs at it with an extra blocker; if the defense goes to base, the tight ends run routes against linebackers. So defenses looked for players who are big enough to stay on the field against a run and fast enough to cover, which is exactly what the big nickel is for. The arms race in one sentence: offenses got smaller to beat base defenses, defenses got smaller to match, and offenses got bigger again to punish the smaller defenses.

! Why it exists

Nickel exists because the passing game won. The rules of the late 1970s made passing far easier (in 1978 the NFL stopped defenders from contacting receivers more than five yards past the line, and let offensive linemen pass-block with extended arms and open hands), and offenses answered with more receivers on the field (The 1978 Liberation tells that story).⁷ Once the third receiver is on the field for most snaps, a package with a linebacker covering him is a package with a weakness on most snaps. **What nickel gives up** is size: about 50 pounds of run-stopper replaced by a cover player. That is why the rise of 12 personnel in 2025 pushed some defenses back toward base, and why the players who can be both (the big nickel safety, the cover linebacker) are among the most valued defenders in the league.

i Film room: Seattle Seahawks 2025

Seattle's Super Bowl LX defense is the cleanest recent example of nickel as the true base. Next Gen Stats found that the Seahawks kept their nickel personnel on the field for 77.8 percent of snaps against offenses using fewer than three wide receivers, the heavier groupings that usually pull a defense into base; no other team was above 60.5 percent. The key was a rookie, Nick Emmanwori, a 6-foot-3, 220-pound defensive back whose size, in Next Gen Stats' phrase, gives Seattle's nickel "the strength of a base unit" without giving up its speed.⁸

What to notice: when a two-tight-end offense comes on against Seattle, watch the substitution. The linebackers do not run on. Instead, find the biggest defensive back on the field and watch where he lines up: often close to the line, where a linebacker would stand. That is the big-nickel idea in practice: the defense answers size with a defensive back who can play like a linebacker, rather than with a linebacker.

The box: counting the run defenders

The next question is where the defenders stand, and the first number every analyst checks is the box count. The **box** is the area near the ball where the run is defended: across, from just outside one end of the offensive line to just outside the other (including a tight end who is attached to the line), and deep, from the line of scrimmage back about 5 to 7 yards. The **box count** is how many defenders are standing inside it at the snap. Data providers draw the box slightly differently,⁹ so treat the edges as fuzzy; the idea is "the defenders close enough to the ball to take on a run in the first second."

Why count them? Because a running play is, at its simplest, an arithmetic problem. The offense has a certain number of blockers: the five linemen, plus any tight end or fullback who stays to block. The runner is not a blocker, and on a normal handoff neither is the quarterback. If the defense has as many defenders in the box as the offense has blockers, every defender can be blocked, and the run is a fair fight, one blocker per defender. If the defense has one more, there is a defender nobody can block, and he is the one

25% in any of the preceding five seasons, dropping to a low of 20.5% in 2023": <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

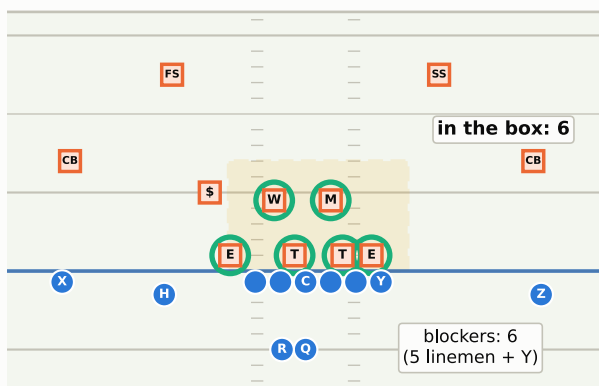
⁷NFL Football Operations, "Evolution of the NFL Rules": <https://operations.nfl.com/the-rules/evolution-of-the-nfl-rules/>. The 1978 changes are told in full in The 1978 Liberation. Pro Football Hall of Fame, "1978" chronology ("permitted a defender to maintain contact with a receiver within five yards of the line of scrimmage, but restricted contact beyond that point"; "The pass-blocking rule was interpreted to permit the extending of arms and open hands"): <https://www.profootballhof.com/football-history/football-history/1960-1979/1978>.

⁸Same NFL.com Next Gen Stats article as the base-rate figures above: Seattle left its nickel personnel on the field "for 77.8% of plays against offensive sub packages – no other team this season was above 60.5%"; Emmanwori's size "gives the Seahawks' nickel defense the strength of a base unit, while allowing it to remain as fast and athletic as any other team's five-DB grouping." In that article "sub packages" means offensive groupings with fewer than three wide receivers.

⁹Next Gen Stats publishes the share of a runner's carries against "8+ defenders in the box" computed from player positions at the snap; nflverse documents the field ([percent_attempts_gte_eight_defenders](#)) in its Next Gen Stats data dictionary: https://nflreadr.nflverse.com/articles/dictionary_nextgen_stats.html ("On every play, Next Gen Stats calculates how many defenders are stacked in the box at snap"). Neither NGS nor nflverse publishes the exact lateral and depth limits it uses. Charting services (FTN, PFF) count the box by eye, with their own rules for edge players.

who makes the tackle. Coaches call this hat math (“a hat on a hat”) and the run-game chapters use it constantly (The Language of the Run Game).

Two deep safeties: 6 in the box



A safety walks down: 7 in the box

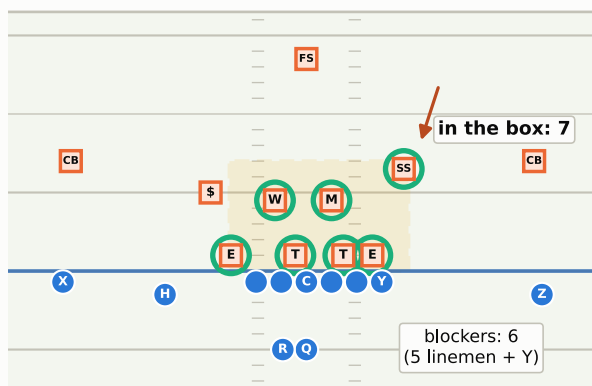


Figure 4: Counting the box. Same offense (11 personnel: five linemen plus the tight end, Y, make six blockers) and the same nickel package. Left: both safeties deep, so the box (shaded: one step outside the left tackle to one step outside the tight end, about 7 yards deep) holds six defenders, ringed: four linemen and two linebackers. Six blockers against six: a fair fight. Right: the strong safety (SS) walks down beside the tight end. Now seven are in the box against six blockers, so one run defender cannot be blocked. The nickel back (\$) stands just outside the box; you will meet him again in a moment.

Two words describe which side the arithmetic favors. A **light box** has no more defenders than the offense has blockers, so every one of them can be blocked; against the usual one-back offense with six blockers, that means six or fewer, and five is very light. A **loaded box** has more defenders than the offense can block, usually eight or more (“eight in the box”), and the extra man almost always comes from the secondary: a safety walking down from 12 yards to 7. Seven sits in between. Against six blockers it is already one too many, but it is so common in the NFL that analysts treat it as the normal box and save “loaded” for eight.

Now the key idea: **the box is a budget**. The defense has eleven players. If seven are in the box and four rush or cover near the line, there are not many left to guard the deep field. Every defender added to the box is a defender subtracted from pass coverage, and every safety kept deep is a defender subtracted from the run. That is why the box count is the first number an offense checks before deciding between run and pass, and why a quarterback with the freedom to change the play at the line (a check or kill call) will often do it by counting the box. Light box: keep the run. Loaded box: throw, because the defense has spent the players who would cover.

The data shows both sides of the bargain, and one trap.

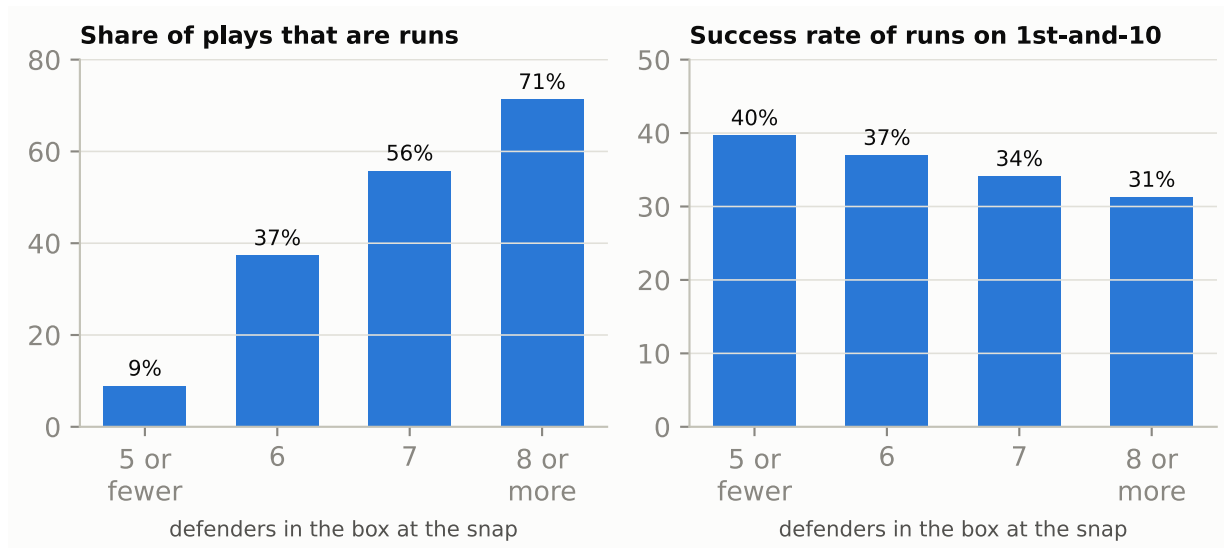


Figure 5: Box count and play choice, regular seasons 2023-2025 pooled. Left: the more defenders in the box, the more often the play is a run, because defenses load the box when they expect a run (short yardage, two-tight-end personnel) and empty it when they expect a pass. Right: on first-and-10, where the situation is the same on every play, runs into heavier boxes succeed less often. Success means the play gained enough to keep the offense on schedule (EPA above zero; see Downs, Distance, and Field Position).

Plays with no box count recorded are left out. Data: nflverse play-by-play and FTN participation via nflreadpy.

The left panel is the defense reading the offense: boxes get heavier as runs get likelier, from under one play in ten being a run against five or fewer in the box to about seven in ten against eight or more. The right panel is the cost of running into a crowd. On first-and-10, a neutral down where nobody is forced into anything, runs into a box of five or fewer succeeded about 40 percent of the time and runs into eight or more about 31 percent. Average yards per carry tell the same story more loudly: about 5.5 yards against the lightest boxes, a little over 3 against loaded ones.¹⁰

The trap is to compare runs across all downs without asking *when* each box happens. Pool every down and the success rates by box count come out almost flat, because loaded boxes show up on third-and-1 and at the goal line, where a 2-yard run counts as a success, and light boxes show up on third-and-9, where it doesn't. The box count and the situation travel together. Holding the situation fixed, as the first-and-10 panel does, is what lets the cost of a loaded box show through. That kind of selection effect is everywhere in football data, and Expected Points and Success Rate deals with it head on.

🔥 Common misconception

"Six in the box? Easy: run the ball." Sometimes. But a light box is a choice the defense made on purpose, usually because it is more afraid of the pass, and it often comes with two deep safeties who can still run downhill to the ball after the snap. The defense is saying "we will give you 4 yards on the ground; we will not give you 40 in the air." Whether taking the 4 yards is smart depends on down and distance: on second-and-5 it is a gift, on third-and-11 it is a punt. And defenders can move: a safety who is 12 yards deep at the snap can be in the box a second later (that move, a rotation, is a story for Disguise and Rotation).

The apex and the overhang defender

Look again at the box diagram. The nickel back is not in the box, but he is not out with the receivers either. He is standing halfway between the left tackle and the slot receiver, a few yards off the ball. That spot has a name. A defender who splits the difference between the last man in the box and a slot receiver is in an **apex** alignment ("he's apexed"; the apex alignment), and the defender who lives there, outside the box but inside the slot, is the **overhang defender**, or overhanger. In a nickel defense he is usually the nickel back; in base, an outside linebacker; in some calls, a safety.

¹⁰This book's calculation from nflverse play-by-play joined to FTN participation, regular seasons 2023-2025 pooled, plays with a recorded box count. Runs: about 5.5 yards per carry against five or fewer in the box, 4.5 against six, 4.1 against seven and 3.3 against eight or more. Pooled over all downs, run success rates were 42, 41, 40 and 42 percent, flat because the average distance to go falls from about 9.5 yards to about 6 as the box gets heavier.

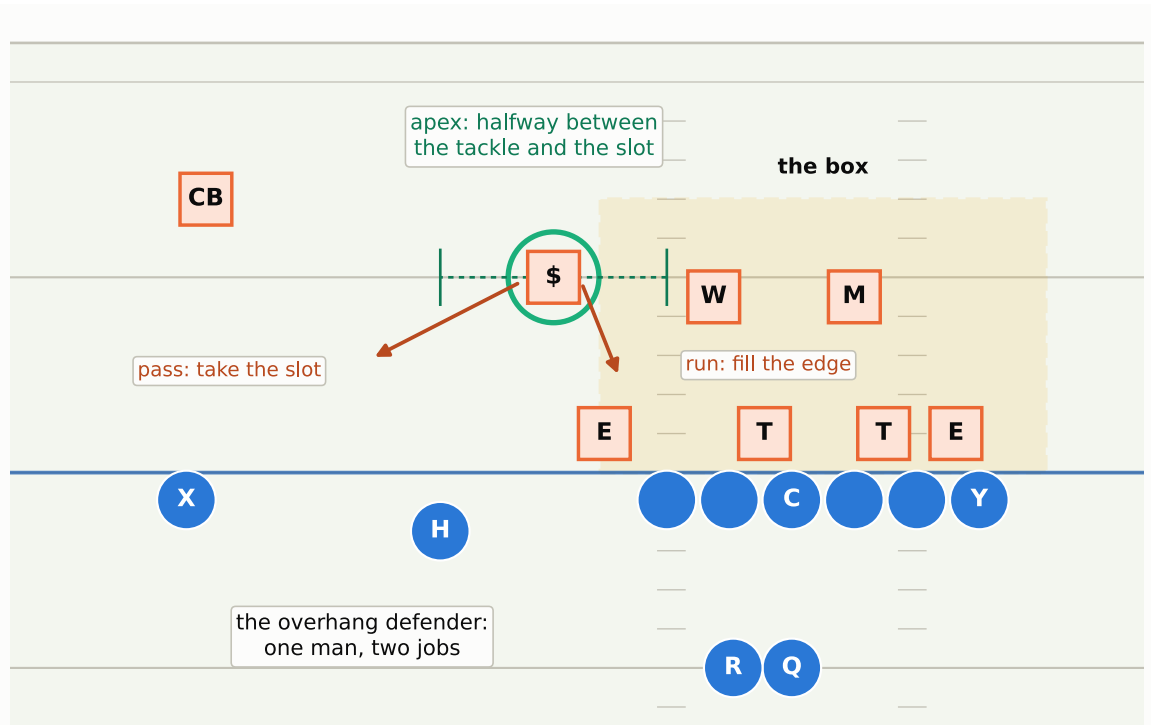


Figure 6: The apex (the offense's left side, zoomed in). The nickel back (ringed) lines up halfway between the left tackle, the edge of the box, and the slot receiver, H. From there he has two jobs that pull in opposite directions: if the play is a run, he must come in and fill the edge of the box; if it is a pass, he must get out and take the slot receiver. He cannot do both at once, which is why offenses build plays to make him choose.

Why stand in such an awkward place? Because both of the alternatives are worse. If he lines up inside the box, he is a strong run defender and the slot receiver is uncovered for a quick throw. If he lines up on the slot receiver, the slot is covered and the box is one man lighter. The apex is the compromise that keeps both threats honest for a split second, and the price is that he is a step late to whichever one actually happens.

That makes the overhang defender the most stressed player on many defenses, and offenses know exactly where he stands. A large part of modern offense is built to put him in conflict, to make his two jobs happen at the same moment so that whatever he does is wrong. You will see him as the conflict defender in Passing Game Fundamentals, as the man the quarterback reads on the run-pass option in RPOs, and from the defense's side in Linebackers and the Second Level and Run Fits and Box Math. For counting purposes, analysts usually leave an apexed defender out of the box count. In your head, count him as half: he can arrive late to a run.

Man and zone: two ways to guard receivers

Now the pass. Once the ball is snapped and receivers start running, every defender who is not rushing has to know which receiver, or which piece of grass, is his. There are two basic answers, and every coverage in Part 6 is built from them.

In **man coverage**, each defender is assigned one eligible receiver and follows him wherever he goes: across the field, deep, back toward the line. In **zone coverage**, each defender is assigned an area of the field and guards whichever receiver comes into it, usually with his eyes on the quarterback rather than on any one receiver. The simplest way to see the difference is to run the same routes against both.

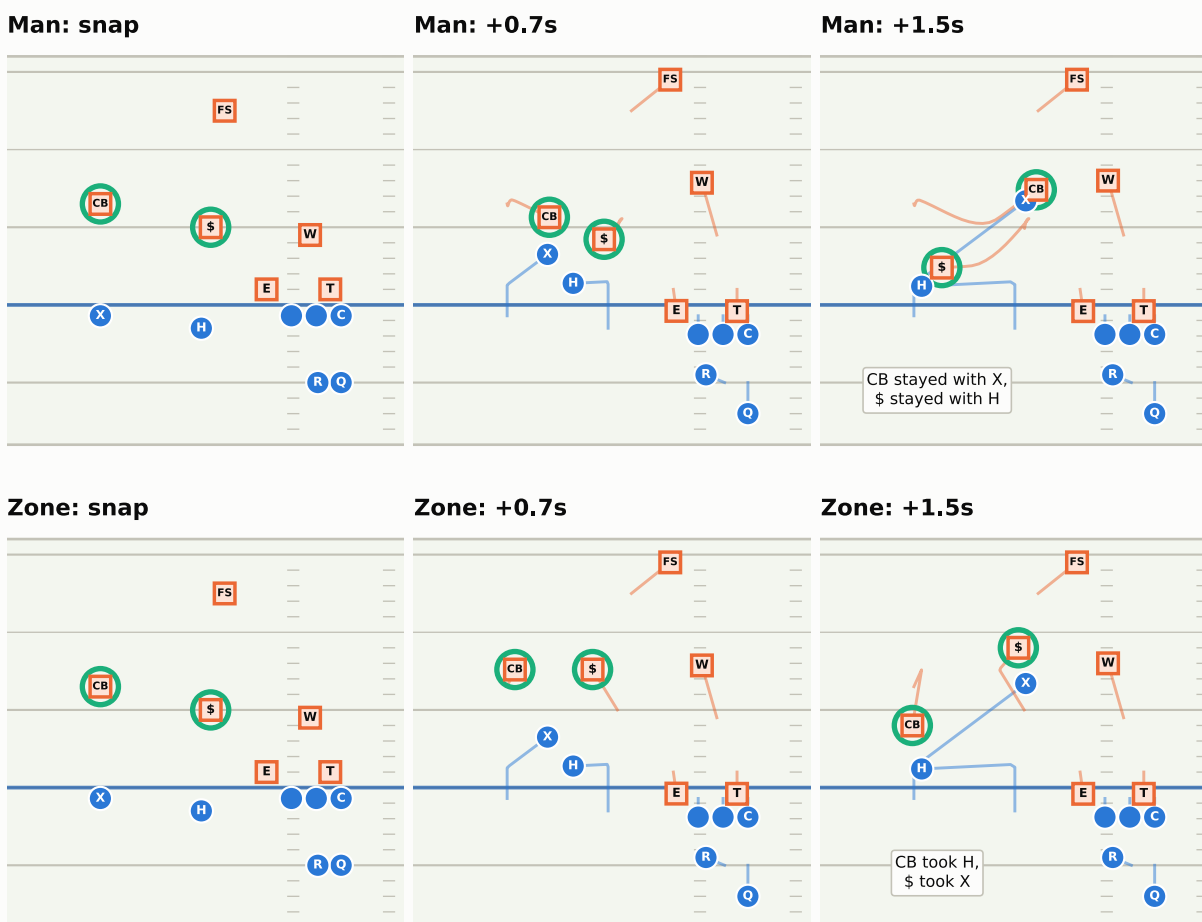


Figure 7: The same two routes against man and against zone (the offense's left side; simulated play). X runs a slant (a quick angle inside), H a shallow out toward the sideline, so the routes cross. Top row, man: the cornerback (CB) is assigned X and the nickel back (\$) is assigned H; each follows his receiver, so the two defenders cross too and finish on the receivers' hips, with their backs to the quarterback. Bottom row, zone: the cornerback guards the short area near the sideline and the nickel back the area inside it, about 8 to 10 yards deep; each sinks into his area watching the quarterback and takes whoever arrives, so they finish on the opposite receivers from the top row. Faint lines are where each player has been. In the web version this is a video.

Same routes, same starting spots, different finish. Against man, the defenders' paths look like the receivers' paths, slightly behind. Against zone, the defenders hardly move until a receiver comes to them, and they end up covering the receivers they were *not* across from at the snap. Once you see that, you can often call man or zone on a replay: **did the defender run with a receiver, or did he run to a spot and wait?**

Each has a clear set of strengths and a clear price.

Man coverage is tight. A good cornerback in man can stay on a receiver's hip and contest every throw, which makes it strong against quick, precise passes and lets a defense put its best cover player on the offense's best receiver. The costs: each defender has his back to the quarterback, so a quarterback who scrambles runs past defenders who are looking the other way; routes that cross or run into each other can pick off a chasing defender, the way a screen does in basketball; and if one defender loses his man, there is often nobody behind him to help. Man coverage is also how most blitzes are played, because sending extra rushers leaves just enough defenders to take one receiver each.

Zone coverage keeps everyone's eyes on the quarterback. Defenders can react to the throw, break on the ball, and rally to tackle, and a scrambling quarterback runs into players facing him. Zone is also safer against the big play, because receivers are passed from one defender to the next rather than chased. The cost is the seams. Every zone has edges and holes where one area ends and the next begins, and good passing offenses are largely a catalog of ways to put a receiver in those gaps, or to put two receivers in one defender's area at once so that he can only guard one (Coverage Fundamentals draws the zones).

NFL defenses play more zone than man. In the NGS-charted seasons, roughly two-thirds (63 to 71 percent) of labelled dropbacks were zone; the charting source changed in 2023 and the man share jumps around after that, which looks like a change in how

plays are labelled rather than in how teams defend, so treat any single recent season with caution.¹¹ And the line between them is blurrier than this section makes it sound. Most modern zones “match” routes: a zone defender who sees a receiver enter his area and stay there starts to cover him like man (pattern matching, in Quarters and Match). For a first look, man versus zone is the right first cut.

You will also start hearing coverages called by numbers: Cover 0, Cover 1, Cover 2, Cover 3, Cover 4. Those numbers mostly count the deep defenders, and Cover 0 and Cover 1 are the main man coverages while Cover 2, 3 and 4 are the main zones (the full system, with its exceptions, is in Coverage Fundamentals). Which brings us to the deep defenders.

One-high and two-high: reading the shell

The single most useful thing you can see before the snap is how many safeties are deep. The safeties are the defense’s last line; where they stand tells you which part of the field the defense is most afraid of. The picture they show before the snap is the **shell**, and there are two basic shells.

- **One-high** (also single-high): one safety deep in the middle of the field, usually 12 to 15 yards off the ball, and the other safety down closer to the line. Coaches say the middle of the field is “closed” (MOFC, middle of field closed).
- **Two-high** (also split-safety): two safeties deep, one on each side of the middle, usually 10 to 14 yards off the ball and a few yards outside the hash marks. The middle of the field between them is “open” (MOFO).

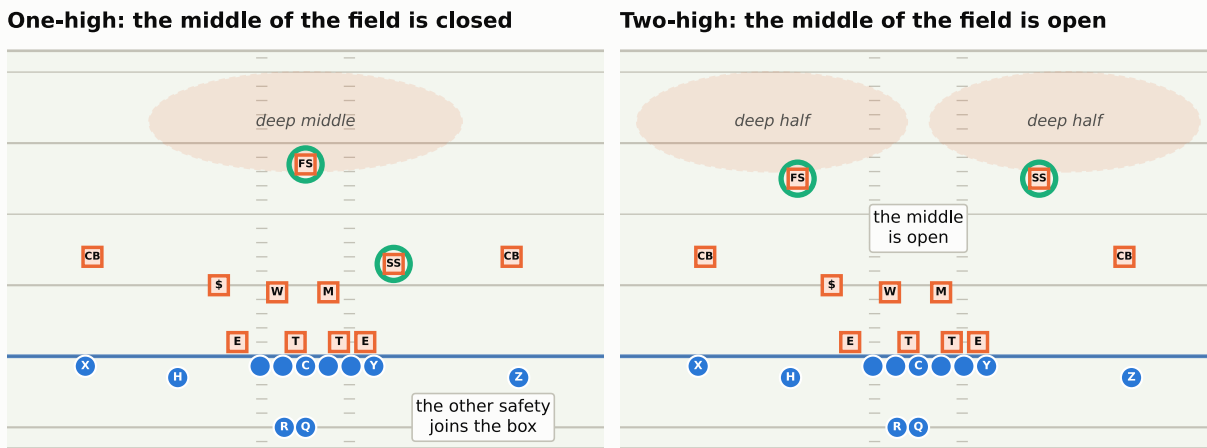


Figure 8: The two shells against the same offense (both safeties ringed). Left, one-high: the free safety (FS) sits deep in the middle, so the deep middle is covered (closed), and the strong safety (SS) is down at about 7 yards beside the tight end, an eighth defender close to the run. Right, two-high: both safeties are deep, each guarding a deep half of the field; between them the middle is open, and both are far from the run. The shaded ovals show the deep areas each shell protects.

Each shell is a different way of spending the same two players, and each is good at something specific.

One-high is built to stop the run and to challenge receivers. With one safety guarding the whole deep middle, the second safety is free to come down, which is how a defense gets seven or eight into the box against a run without leaving the deep field empty. One-high is also the natural home of man coverage (Cover 1: man on everyone, one safety deep as the insurance) and of the most common zone, Cover 3 (three deep, four under). Its weakness is width: one man in the middle cannot help both sidelines at once, so the deep outside thirds and the seams between them are where offenses attack.

Two-high is built to stop the big pass. With a safety over each half of the field, a deep throw to either sideline has a defender coming over the top, and that safety can also squeeze down on a crossing route in front of him. The price is the run: both safeties start 12 yards away, so the box is lighter (usually six), and the middle of the field between them, short and intermediate, is open for throws.

So the shell and the box are two readings of the same budget. **One-high usually comes with a heavier box; two-high usually comes with a lighter one.** If you count seven in the box, look for the second safety; you will usually find him at 7 yards, not 12.

¹¹This book’s calculation from nflverse participation data (`defense_man_zone_type`): zone was 65 percent of labelled plays in 2018, 70 percent in 2021 and 71 percent in 2022. After the 2023 switch to FTN charting the man share of labelled dropbacks reads 41 percent (2023), 48 percent (2024) and 31 percent (2025), jumps that look like changes in labelling rather than in play.

That is also why the offense's best guess before the snap often comes from the safeties: two-high and six in the box says "we're daring you to run," and one-high with eight down says "we're daring you to throw."

! Why it exists

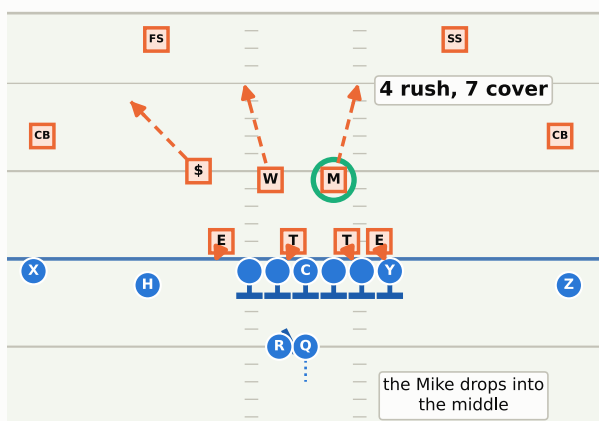
The rise of two-high. For most of the last decade, NFL offenses got better at the explosive pass, and defenses answered by keeping two safeties deep more often. Next Gen Stats' split-safety rate went from 32.9 percent in 2018 to 37.8 percent in 2021, 40.4 percent in 2024 and 42.0 percent in 2025, the highest since its tracking began; in 2025 every defense was above 30 percent.¹² **What it gives up** is the run, which is why the same years brought the renewed interest in two-tight-end sets and wide-zone running you will meet in Part 3: if the defense is going to give you six in the box, run at it with seven blockers. The case study From Two-High to Macdonald follows the whole cycle.

One warning before you trust the shell completely. The safeties know you are looking at them. A defense can show two-high and rotate to one-high at the snap, or the reverse, precisely to make the pre-snap read wrong; it is called disguise and it has its own chapter, Disguise, Rotation, and Split-Field Coverage. Read the shell anyway. It is a clue, not a promise, and most of the time it is honest, because a safety who has to sprint 8 yards at the snap is late to wherever he is going.

Four-man rush and the blitz

The last question is how many defenders go after the quarterback on a pass. The default is the **four-man rush**: four players rush, usually the four linemen (or, in a 3-4, three linemen and an outside linebacker), and the other seven cover. A **blitz** is a rush with five or more, which means at least one player who would normally cover, a linebacker or a defensive back, comes after the quarterback instead. On the broadcast the count is the thing: who the four are can vary; *how many* rush is what makes it a blitz.

Four-man rush



Blitz: the Mike rushes too

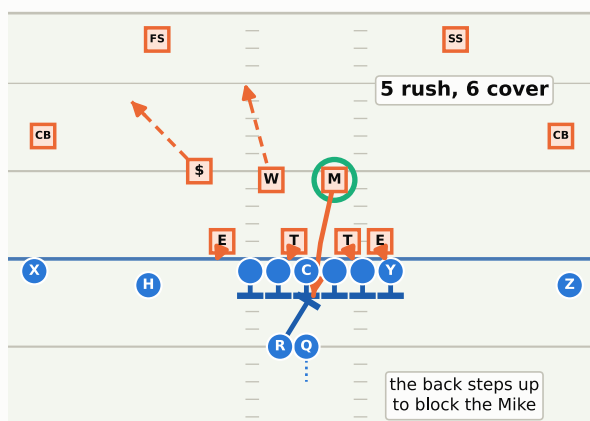


Figure 9: The same pre-snap look, two different plays. Left: a four-man rush. The four linemen rush into the five offensive linemen; the Mike (ringed), Will and nickel back drop into coverage (dashed), so seven defenders cover. The tight end has stayed in to block here as well. Right: a blitz. The Mike rushes through the gap beside the center, making five rushers and leaving six to cover. The offense has a sixth blocker, the running back, who steps up to meet him. If the back had gone out on a pass route instead, the Mike would have been unblocked.

Why is four the default? Because of the arithmetic on the other side. The offense always has five linemen blocking on a pass, and often a back or tight end as well. Four rushers against five or six blockers will rarely get a free run at the quarterback, so the rush has to win with skill: speed, power, and moves that beat a blocker one-on-one (Pass Rush). In exchange, seven defenders cover five possible receivers, which leaves room for two safeties deep and for doubling a star receiver.

A blitz flips the trade. With five or six rushers, the defense tries to create a rusher the offense cannot block, a free man, who reaches the quarterback before the play can develop. The cost is coverage: with six or five defenders left to cover, there is less help deep and fewer people to tackle after a catch. If the offense blocks everyone, or the quarterback gets rid of the ball quickly to the spot

¹²NFL Next Gen Stats Analytics Team, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm," NFL.com (2026), split-safety rate by season and "every single defense exceeded the 30% threshold in 2025": <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>.

the blitz just left (a hot route, in Pass Protection), a blitz can give up a big play. Counting rushers against blockers, and working out who is free, is “blitz math,” and Blitzes and Man Pressure does it properly.

How often do teams blitz? Less than television makes it feel, and the reason shows up as soon as you put the two sides of the trade next to each other.

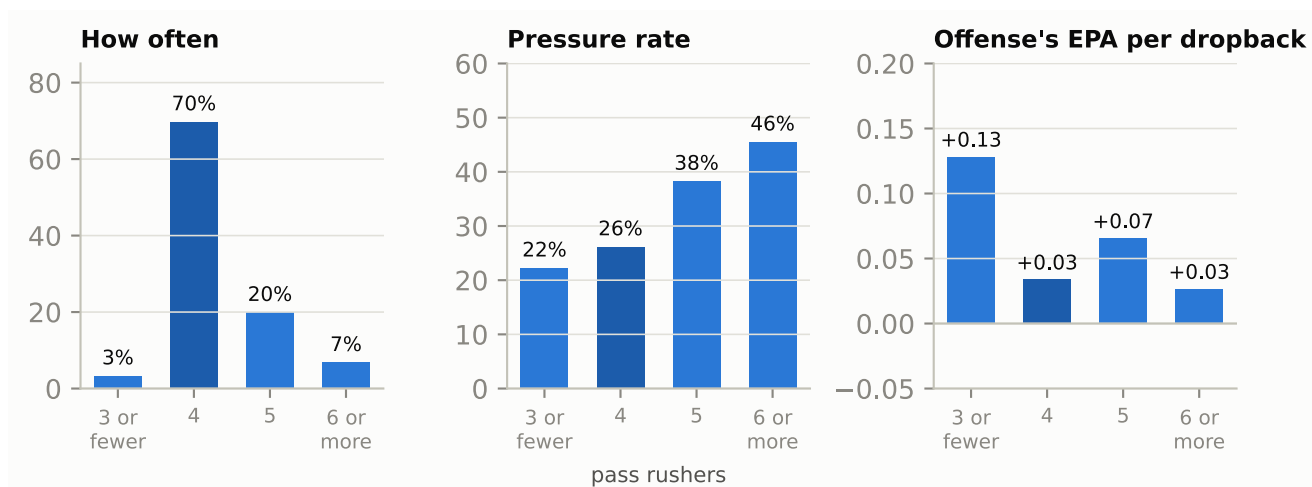


Figure 10: What extra rushers buy, 2025 regular season (all dropbacks with the number of rushers charted, sacks and scrambles included). Left: how often the defense rushed three or fewer, four, five, or six or more. Middle: the share of those dropbacks on which the quarterback was pressured (hurried, hit or sacked) rises steadily with each extra rusher. Right: yet the offense’s average result per dropback, in expected points added (EPA, introduced in Downs, Distance, and Field Position; higher is better for the offense), barely changes, because every extra rusher is a defender taken out of coverage. Blitzes are also called more often on third downs and against particular offenses, so the right panel mixes situations; it shows that blitzing is no free lunch, not that it never works. Data: nflverse play-by-play joined to FTN participation charting via nflreadpy.

About seven dropbacks in ten saw exactly four rushers in 2025, a bit more than a quarter saw five or more, and only a few percent saw three or fewer.¹³ The middle panel is why coordinators blitz at all: each extra rusher makes pressure more likely, from about one dropback in four with four rushers to more than four in ten with six. The right panel is why they don’t do it all the time. Pressure is not the goal; stopping the offense is, and on average the offense did about as well against five or six rushers as against four. The pressure came, and so did the open receivers behind it. Charting services count rushers slightly differently, so take the exact figures lightly; the shape, four as the norm and five or more as the change-up, is the same in every source.

Two caveats will save you from the most common mistakes on a broadcast. First, **a linebacker rushing does not make it a blitz**; the count does. In a 3-4, an outside linebacker rushing is the normal four-man rush. Second, defenses love to *show* blitz and not send it (linebackers creeping up into the gaps before the snap and then dropping out), and to send four rushers that include a linebacker while a lineman drops into coverage. That last trick looks like a blitz to the quarterback and is still a four-man rush; it is called simulated pressure, and it gets its own chapter, Zone Blitzes and Simulated Pressure.

¹³This book’s calculation, the chart above: nflverse play-by-play joined to FTN participation charting (`number_of_pass_rushers`, `was_pressure`), 2025 regular season, dropbacks with at least one rusher recorded (about 19,700). Four rushers 69.8 percent, five or more 26.9 percent, three or fewer 3.4 percent. In the NGS-sourced participation data for 2016-2022 the five-or-more share was between 26 and 30 percent every season.

The best recent argument for the four-man rush came on the biggest stage. In Super Bowl LIX (February 9, 2025, the last game of the 2024 season) the Eagles beat the Chiefs 40-22, and Vic Fangio's defense blitzed Patrick Mahomes zero times on his 42 dropbacks, according to Next Gen Stats. They pressured him 16 times anyway and sacked him six times, the most he had ever been sacked in a game.¹⁴

What to notice: on a replay of almost any Kansas City dropback from that game, count the rushers: four, every time. Then count the defenders in coverage: seven, often with two safeties deep. The Eagles' linemen won their one-on-one blocks, so the defense got the benefit of a blitz (pressure) without paying for one (fewer men in coverage). That is the dream every defensive coordinator chases, and it is only possible with linemen who can win alone. When a defense can't get there with four, it has to blitz, and that is where Part 7 picks up.

Putting it together: the five-question read

Every idea in this chapter is the same idea seen from a different angle: eleven defenders, one budget. The package decides what kind of players are available; the box count and the apex say how many are spent on the run; the shell says how many are spent on the deep pass; the rush says how many are spent on the quarterback. Because the total is fixed, the readings agree with each other far more often than not:

Table 2: The pre-snap read of a defense, as far as this chapter takes it.

Question	What you look for	What it suggests
Package	Substitutions; count the defensive backs when you can	Nickel: the normal look. Base: the defense expects a run. Dime: it expects a pass.
Box	Defenders within about 7 yards, tackle to tackle (or tight end)	Equal to the blockers or fewer: the run is fair. More: the run is outnumbered.
Apex	The defender halfway between the box and the slot	The man the offense will try to put in conflict.
Shell	Safeties deep: one or two	One-high: heavier box, more man and blitz. Two-high: lighter box, protects the deep pass.
Rush	Count rushers after the snap	Four: the norm. Five or more: a blitz, and coverage behind it is thin.

None of these readings is certain. A box can be loaded and still drop eight into coverage; a two-high shell can rotate; a linebacker can show blitz and drop. Later chapters are, in a sense, a list of the ways defenses make these clues lie, and the ways offenses force the truth out, starting with the next chapter: Motion and Shifts as Weapons, whose main job is to make the defense show what it is doing.

Film room

Two defenses, thirty-five years apart, that spent their eleven players on purpose.

¹⁴Rohan Nadkarni, "The Eagles made Patrick Mahomes look mortal," NBC News, Feb 10, 2025 ("blitzed Mahomes a grand total of zero times on his 42 dropbacks"; "pressured Mahomes 16 times"; six sacks, "the most he has ever been sacked in a game in his career"): <https://www.nbcnews.com/sports/nfl/eagles-made-patrick-mahomes-look-mortal-rcna191414>.

i Film room: New York Giants 1990

The most famous light box in NFL history was on purpose. Facing the Buffalo Bills' fast, pass-heavy no-huddle offense in Super Bowl XXV (January 27, 1991, the end of the 1990 season), Giants defensive coordinator Bill Belichick built a game plan around defensive backs: the Giants started the game with extra defensive backs on the field and at times only two down linemen, in looks Bill Parcells later described as "a 2-4-5, or a 2-3-6." Belichick told his players they had to let running back Thurman Thomas run for 100 yards; as he explained, he did not want a game in which Buffalo threw 45 times, and the extra defensive backs were meant to "entice them into calling more runs." Thomas ran for 135 yards. The Giants won 20-19 when Buffalo's last-second field goal missed, and the game plan is now in the Pro Football Hall of Fame.¹⁵

What to notice: this is the box as a budget, spent deliberately. The Giants gave up the cheap yards on the ground to keep extra defenders in coverage against Buffalo's receivers. A light box is not a mistake; it is a decision about which yards you can afford.

i Film room: Baltimore Ravens 2023–2025

The modern big nickel has a face: Baltimore's Kyle Hamilton, a safety the Ravens drafted in the first round in 2022. In 2023, with Mike Macdonald as defensive coordinator, Baltimore lived in the package built around him. By this book's count of the charted personnel, the Ravens had five defensive backs with three or more of them safeties on about three snaps in four (76 percent), five times the league rate (15 percent), and played base on fewer than one snap in five.¹⁶ Hamilton was the third safety, and where he lined up decided what the defense was that snap. The idea travelled: Macdonald became Seattle's head coach in 2024, and his 2025 Seahawks, built the same way, won Super Bowl LX.

Baltimore kept the package after Macdonald left. In 2025 Hamilton played more snaps over the slot (249) than as a traditional deep safety (228), according to ESPN, and one evaluator in ESPN's annual survey of executives, coaches and scouts put it bluntly: "He's an average safety but he's an elite big nickel, which he's in most of the time."¹⁷

What to notice: before the snap, find number 14. If he is 12 yards deep, Baltimore is showing a traditional two-high look. If he is 5 yards off the ball over the slot receiver or the tight end, he is the overhang defender, and you are watching a defensive back built like a linebacker, able to fill the edge against a run or carry the slot receiver down the field. That flexibility is what lets the defense stay in one package whatever personnel the offense sends on, and it is why offenses spend the week figuring out where he will be.

¹⁵Michael Eisen, "Belichick's gameplan," Giants.com (Parcells: "We played a 2-4-5, or a 2-3-6"; Belichick: "I didn't feel like we wanted to get into a game where they threw the ball 45 times"; the plan "now displayed in the Hall of Fame"): <https://www.giants.com/news/belichick-s-gameplan>; Super Bowl XXV box score (Thomas 15 carries, 135 yards; Giants 20, Bills 19): <https://www.pro-football-reference.com/boxscores/199101270buf.htm>.

¹⁶This book's calculation from nflverse participation data (FTN charting), 2023 regular season, run and pass plays with defensive personnel recorded. "Big nickel" here means exactly five defensive backs, at least three of them listed as safeties (S, FS or SS) on the roster. Baltimore: 75.6 percent of snaps; league: 15.0 percent; Baltimore in four-DB (base) packages: 18.0 percent. Roster labels are an imperfect stand-in for what a player does on a given snap. Seattle 2025, same method: 55.6 percent big nickel, 7.1 percent base. Macdonald was Baltimore's defensive coordinator in 2022-2023 and Seattle's head coach from 2024: Wikipedia, "Mike Macdonald (American football)": [https://en.wikipedia.org/wiki/Mike_Macdonald_\(American_football\)](https://en.wikipedia.org/wiki/Mike_Macdonald_(American_football)). Seattle hired him on January 31, 2024 (AP, "Seahawks expected to hire Ravens defensive coordinator Mike Macdonald as new coach": <https://www.ksat.com/sports/2024/01/31/seahawks-expected-to-hire-ravens-defensive-coordinator-mike-macdonald-as-new-coach-ap-source-says/>). Hamilton was the 14th overall pick in 2022 and wears 14: Baltimore Ravens, <https://www.baltimoreravens.com/news/kyle-hamilton-tyler-linderbaum-david-ojabo-draft-jersey-numbers-2022>. Baltimore after Macdonald, same method: 64.6 percent big nickel in 2024 and 38.2 percent in 2025 (league 15.8 and 18.5). Super Bowl LX (Seattle 29, New England 13, February 8, 2026): https://en.wikipedia.org/wiki/Super_Bowl_LX.

¹⁷Jeremy Fowler, "Execs, coaches, scouts rank top NFL safeties for 2026," ESPN, Jul 16, 2026 (the "elite big nickel" quote, attributed to an unnamed NFL offensive coach; 249 snaps in the slot vs 228 as a traditional safety in 2025): https://www.espn.com/nfl/story/_/id/49307927/ranking-nfl-top-10-safeties-2026-execs-coaches-scouts.

Watch for it on Sunday

Watch for it

On Sunday: box and shell, every snap. The goal is a two-number habit you can do in the five seconds before the snap.

When the players run on

- Did the defense substitute after the offense did? If a big body ran off and a defensive back ran on, it is in nickel or dime. If it stayed big against three receivers, someone in the front seven will have to cover a slot.

Before the snap

1. **Count the box.** Defenders within about 7 yards of the ball, from just outside one end of the line to just outside the other. Then count the offense's blockers: five linemen, plus any tight end on the line and any fullback. Say it as a matchup: "six on six," "seven on six."
2. **Find the safeties.** One deep in the middle, or two deep and split? Say "one-high" or "two-high." If the box count is high, expect one-high; check that the numbers agree.
3. **Find the apex.** Who is standing between the box and the slot receiver? Remember him; he is the player most likely to be wrong.

After the snap

4. **Count the rushers.** Four, or five and more? If five or more, look for the quick throw and the receiver nobody is covering.
5. **Man or zone?** Watch one cornerback. Does he turn and run with his receiver, or settle into an area with his eyes on the quarterback?

For your prediction log (How to Watch a Broadcast): next to each run/pass guess, write the box count, the blockers and the shell, for example "6v6, 2-high: run". After a quarter, check whether the offense ran more often when the box was light, and whether two-high shells really did come with six in the box.

Predict the play

💡 Predict the play

Drill 1. First-and-10 at your own 35. The offense brings on 12 personnel: two tight ends, both in-line, one back behind the quarterback under center, two wide receivers. The defense answers by staying in nickel and keeps two safeties deep. Count the box, count the blockers, and decide: run or pass? What would you expect the defense to change if this happens all game?

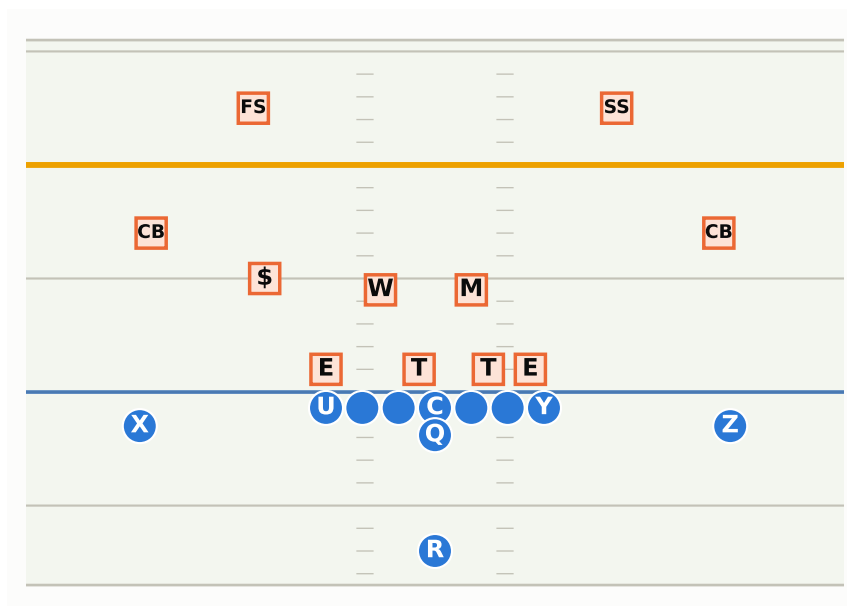


Figure 11: Drill 1: 12 personnel (two in-line tight ends, U and Y) against a nickel defense with two deep safeties. Count the defenders in the box and the offense's blockers.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. Third-and-8 from midfield. The offense is in the shotgun with 11 personnel. Before the snap, both linebackers walk up into the gaps on either side of the center, all three cornerbacks move up to within a yard or two of their receivers, one safety is deep in the middle and the other is down at about 7 yards beside the tight end. How many defenders could rush, how many blockers does the offense have, what kind of coverage do the corners suggest, and what should the quarterback be thinking?

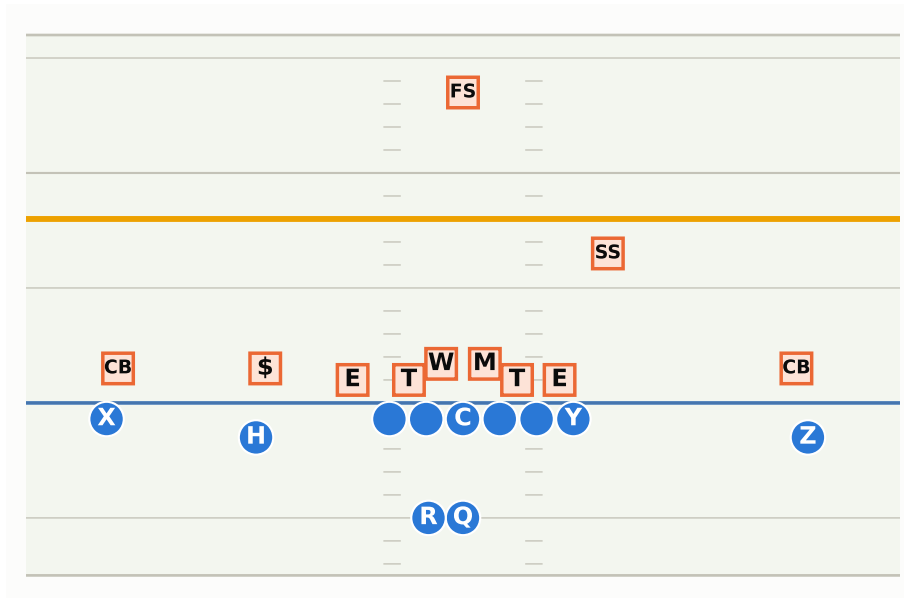


Figure 12: Drill 2: third-and-8. Both linebackers (M, W) stand in the gaps on either side of the center, the corners and nickel back are pressed up on the receivers, one safety is deep in the middle and the other is down beside the tight end.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. Second-and-6. The offense is in 11 personnel with a slot receiver on the left. The defense is in nickel with two deep safeties, six in the box, and the nickel back apexed between the left tackle and the slot receiver. The offense calls a play that looks like an inside run but lets the quarterback throw to the slot instead, depending on what one defender does. Which defender is the quarterback watching, and what does he do if that defender steps toward the line at the snap? If he widens toward the slot?

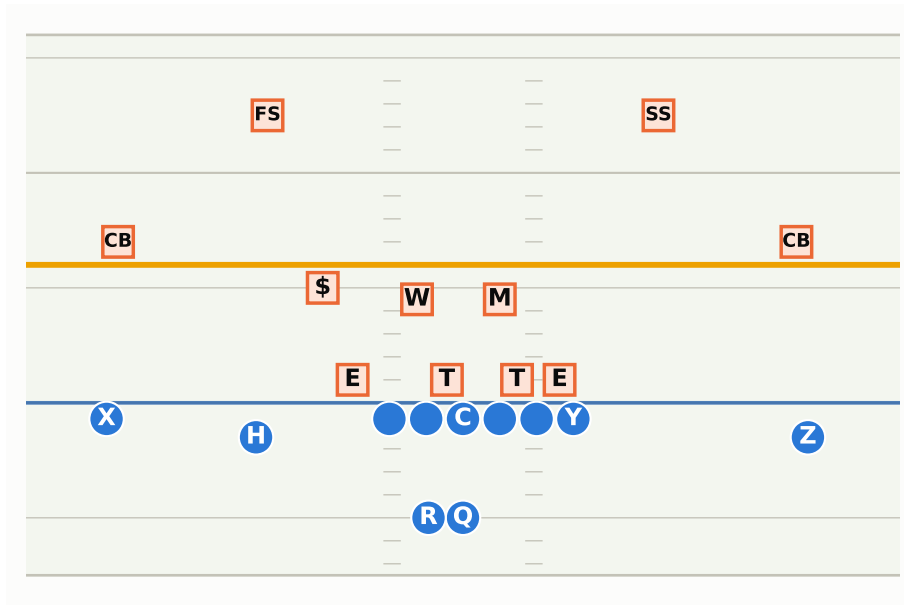


Figure 13: Drill 3: second-and-6, 11 personnel against nickel with two deep safeties. The nickel back (\$) is in an apex alignment between the left tackle and the slot receiver (H). Who is the quarterback reading?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- A defense is a budget: eleven players spent across the run, the deep pass, the short pass and the quarterback. Every pre-snap clue in this chapter is a reading of how that budget was spent, and the readings usually agree with each other.
- **Packages** are named by defensive backs: base (four; a 4-3 or 3-4 front), nickel (five), dime (six), quarter or dollar (seven), with big nickel (a third safety as the fifth DB) and goal line at the heavy end. The defense chooses its package in answer to the offense's personnel. Nickel is now the NFL's real base, because three-receiver offenses are the norm; base packages rebounded in 2025 as offenses went to two tight ends.
- **Count the box and compare it to the blockers.** Equal or fewer, and the run is a fair fight; more, and someone cannot be blocked. A heavier box means fewer defenders in coverage. The **apex** defender between the box and the slot is the half-in, half-out player offenses love to put in conflict.
- **Man** coverage assigns receivers; **zone** assigns areas. Man is tight but blind to the quarterback; zone sees the ball but has seams. NFL defenses play more zone.
- **One-high** (middle of the field closed) pairs with a heavier box and is good against the run and in man coverage; **two-high** (middle open) protects the deep pass and lightens the box. Two-high rose from about a third of snaps in 2018 to 42 percent in 2025.
- A **four-man rush** is the norm; a **blitz** is five or more rushers, trading coverage for a chance at an unblocked rusher. Count rushers, not linebackers.

Connections

Builds on: The Twenty-Two (every defensive position and what it does) and Personnel (the offense's personnel codes and the substitution rules that let the defense match them).

Sets up:

- Motion and Shifts as Weapons: how offenses move players before the snap to make the defense reveal man or zone and its shell.
- The Language of the Run Game: gaps, technique numbers and hat math, the box count turned into blocking assignments.
- Passing Game Fundamentals and RPOs: the conflict defender, and plays built to attack the overhang.
- **Part 5, fronts:** Defensive Line Play, Even Fronts and Odd and Hybrid Fronts (what the 4-3 and 3-4 really mean), and Run Fits and Box Math.
- **Part 6, coverage:** Coverage Fundamentals (zones, the shell read, Cover 0-9), Man Coverage, and Disguise, Rotation, and Split-Field Coverage.
- **Part 7, pressure:** Pass Rush, Blitzes and Man Pressure and Zone Blitzes and Simulated Pressure.
- Appendix C, the defensive atlas, has a one-page card for each package and the 4-3 vs 3-4 vs nickel comparison.

See also (case studies): The Belichick Patriots (game-plan defense, of which Super Bowl XXV was an early example) and From Two-High to Macdonald.

If you want to go deeper: Chris B. Brown's *The Art of Smart Football* (2015) and his Smart Football site on why defenses went to nickel and two-high shells; Tim Layden's *Blood, Sweat and Chalk* (2010) for how defensive systems such as the 3-4 and the zone blitz came to be; and the NFL Next Gen Stats articles on NFL.com, which publish box counts, split-safety rates and blitz rates during the season.

Go deeper

Counting the box in the data. The nflverse participation data (loaded with `nflreadpy`; see nflverse in Python) has a `defenders_in_box` column for every play and a `number_of_pass_rushers` column for most dropbacks, and FTN's charting adds `n_defense_box` and `n_pass_rushers` from 2022 on. Two traps: the source changes between 2022 (Next Gen Stats) and 2023 (FTN), and the two draw the box and list positions differently, so do not splice the two into one trend without marking the break; and some FTN rows record a box of zero, which almost certainly means "not recorded." With real tracking data (the NFL Big Data Bowl files) you can count the box yourself from player positions at the snap, using whatever box definition you like (the code is in the web version of this chapter):

Answers to Predict the play

Answer 1.

Run. The box holds six defenders: four linemen and two linebackers. The nickel back is outside it, near the left tight end, and both safeties are 12 yards deep. The offense has **seven** blockers: five linemen and two tight ends. Seven against six means the offense can block every defender in the box and still have a blocker left over to climb to the nickel back or a safety. This is exactly why offenses turned to 12 personnel in 2025: against a defense that stays in nickel with two deep safeties, the run game has the numbers.

What the defense changes: one of two things. Either it goes bigger with its personnel (base, or a big nickel with a safety who can play near the line), or it keeps its package and brings a safety down before or at the snap, turning two-high into one-high and six in the box into seven. Both cost something in pass coverage, and an offense that sees the safety come down will start throwing play-action passes behind him.

Answer 2.

Six could rush (eight, counting the strong safety and the nickel back), the offense has six blockers (five linemen and the back), the coverage is probably man, and the quarterback should be thinking "get the ball out fast." The linebackers in the gaps beside the center make the offense treat them as rushers, so the back has to stay in to block; if both linebackers come, six rushers meet six blockers with no margin, and if the safety beside the tight end comes too, somebody is free. The corners and nickel back pressed up close on their receivers, with one safety deep in the middle, is the classic picture of man coverage with one deep helper (Cover 1, in Man Coverage). Against that, the quarterback wants a quick throw to a receiver who can beat his man in the first two seconds, before the free rusher arrives.

The catch: this is a *look*, not a promise. Both linebackers might drop back into coverage at the snap, leaving four rushers and seven defenders waiting for the quick throw the quarterback was so sure of. That threat (a defense that can show six and rush four, or show four and rush six) is what Part 7 is about; this two-linebacker look in particular is the double A-gap mug.

Answer 3.

The nickel back, the overhang defender. He has two jobs: fill the edge of the box if it is a run, cover the slot receiver if it is a pass. The play is built so that he cannot do both.

- **If he steps toward the line** (to stop the run), the slot receiver is uncovered for a quick throw, and the quarterback pulls the ball and throws it to him.
- **If he widens toward the slot** (to cover the pass), the box is six defenders against six blockers and the defense has one fewer man near the run, so the quarterback hands the ball off.

This kind of play, a run with a built-in pass that the quarterback chooses after the snap by watching one defender, is a run-pass option, or RPO, and the overhang defender is its favorite target. Defenses answer by moving him around, disguising whether he will play the run or the pass, and by sometimes having a safety do one of his two jobs for him.